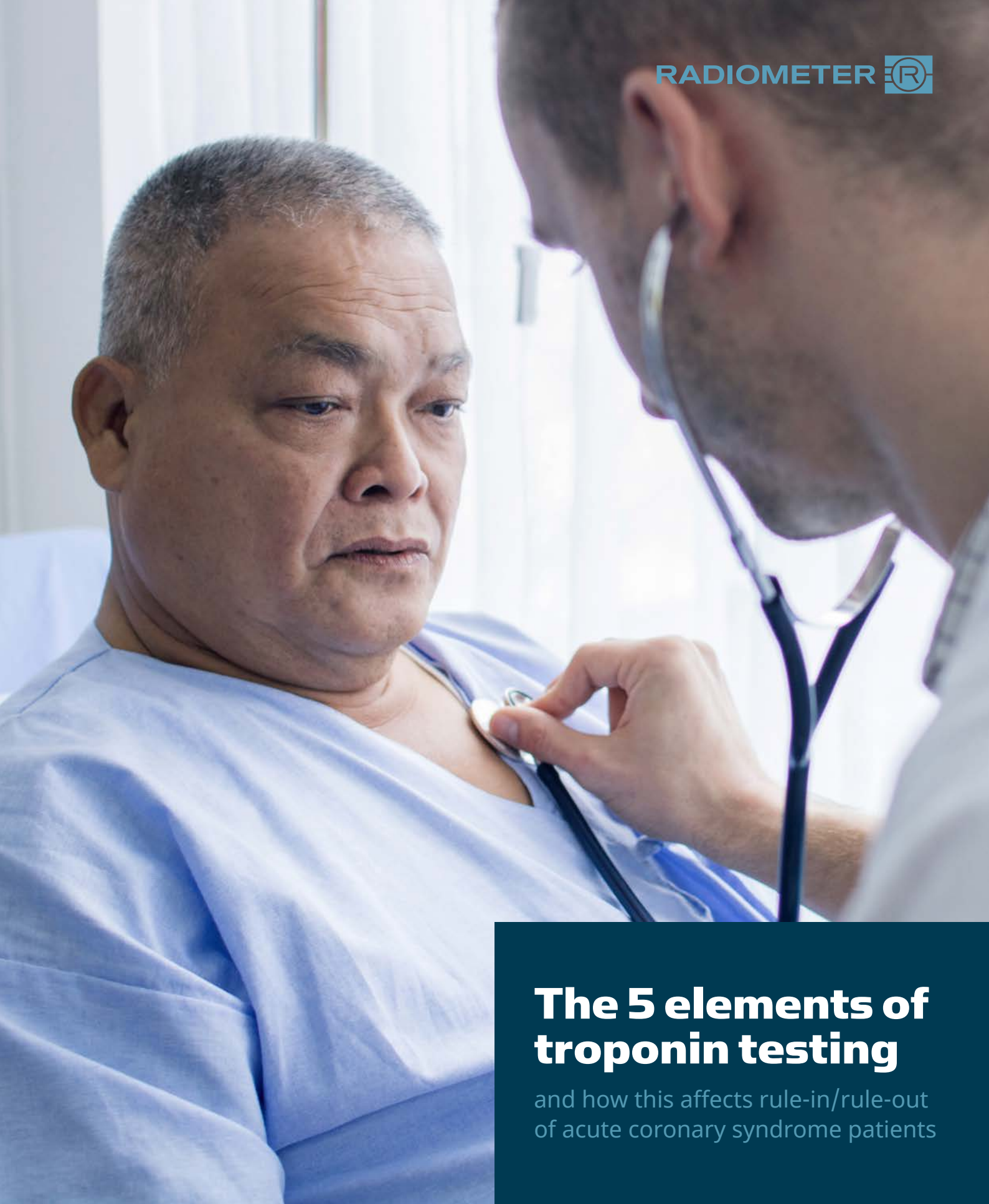


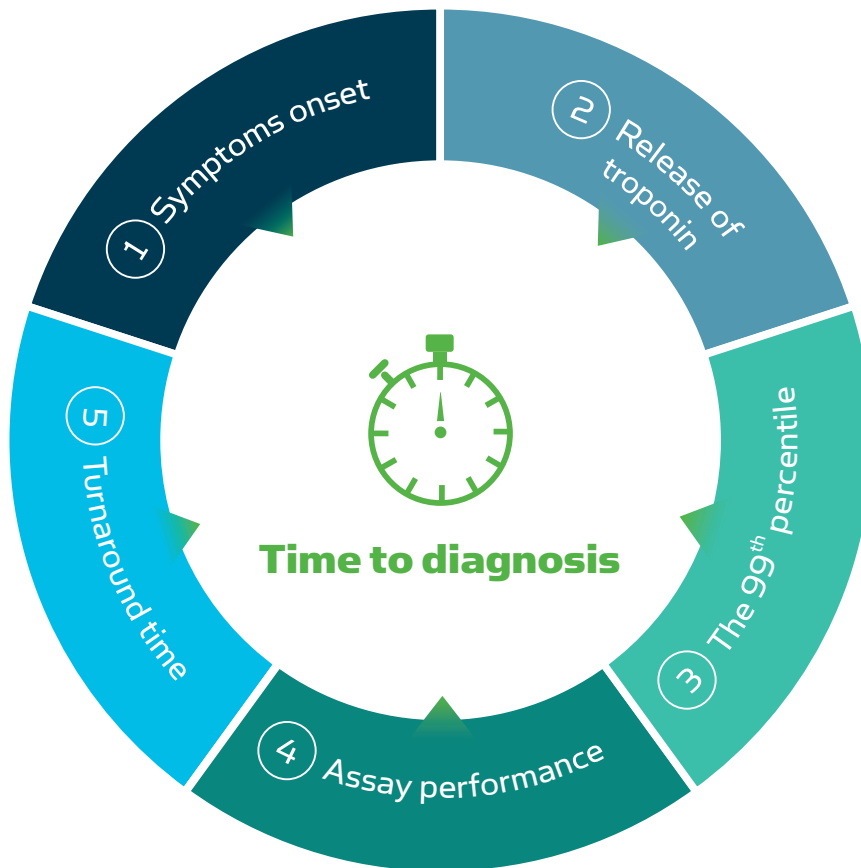


The 5 elements of troponin testing

and how this affects rule-in/rule-out of acute coronary syndrome patients

The five elements of troponin testing

Time to diagnosis [1], the “time protocol” or “time algorithm” is determined by five elements that are all interlinked:



- ① Clinical presentation and symptoms onset, including the delay in presentation to the emergency department (ED)
- ② The dynamic release of troponin into the blood stream and the timing of the sample collections
- ③ The population dependent 99th percentile upper reference limit (URL)
- ④ Analytical and clinical performance of the troponin assay
- ⑤ The turnaround time of the troponin assay

1 Clinical presentation of myocardial infarction (MI)

Myocardial ischemia

Myocardial ischemia arises when oxygen delivery to cardiac tissue is restricted, usually due to reduction in blood flow. Ischemia leads to myocyte necrosis and injury of the heart muscle [2]. Cardiac troponin (cTn) is released into the circulation following ischemia [3].

Clinical symptoms

Some ischemic symptoms are [3]:

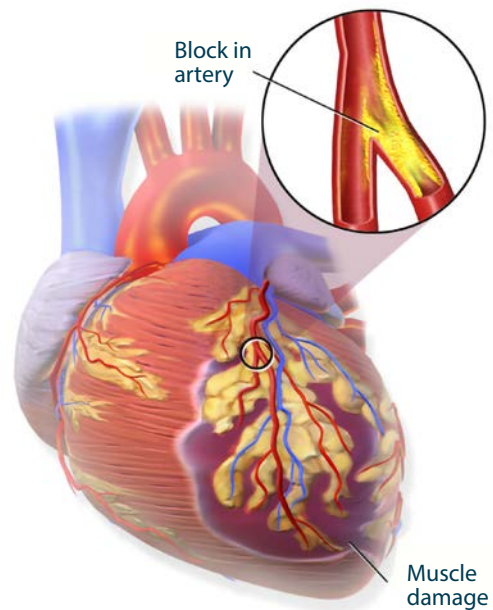
- Pain in chest, left or right arm, neck or jaw
- Epigastric discomfort
- Shortness of breath
- Fatigue

Symptoms may be non-specific, atypical or sometimes there are no symptoms at all but the patient feels unwell [3].

Universal definition of MI [3]

Detection of a rise and/or fall of cTn values with at least one value above the 99th percentile URL and with at least one of the following:

- Symptoms of acute myocardial ischemia
- New ischemic ECG changes
- Development of pathological Q waves
- Imaging evidence of new loss of viable myocardium or new regional wall motion abnormality in a pattern consistent with an ischemic aetiology
- (For type 1 MI) Identification of a coronary thrombus by angiography (including intracoronary imaging) or by autopsy



Symptom onset

Typically, MI patients present to the ED after having chest pain or other symptoms.

Early presenters: patients presenting to the ED less than two hours after symptom onset are designated as “early presenters” [3]. These patients would have to be observed in the ED for a longer time before a clear rise/fall pattern of troponin can be detected [4].

Late presenters: patients presenting to the ED more than two hours after symptom onset [3]. These patients exhibit a clear rise/fall pattern of troponin in case of MI.

2 Dynamic release of troponin

Timing of sample collection

When troponin is released into the blood stream, it is very individual how fast the concentration increases to measurable concentrations.

The concentration of troponin depends on [3]:

- Time from onset of myocardial injury
- The size of the infarct
- Blood flow through the infarcted area. If the vessel is completely blocked (a STEMI) there will be no blood flow but if it is only partially blocked, washout of troponin will occur earlier. In rare cases, patients in shock will present with low troponin values due to low blood flow
- The troponin assay used

It should therefore be appreciated that the time before a troponin result increases above the cut-off (the 99th percentile) and the time before a changing pattern between serial measurements is recognized will vary between patients [3].

Time protocol

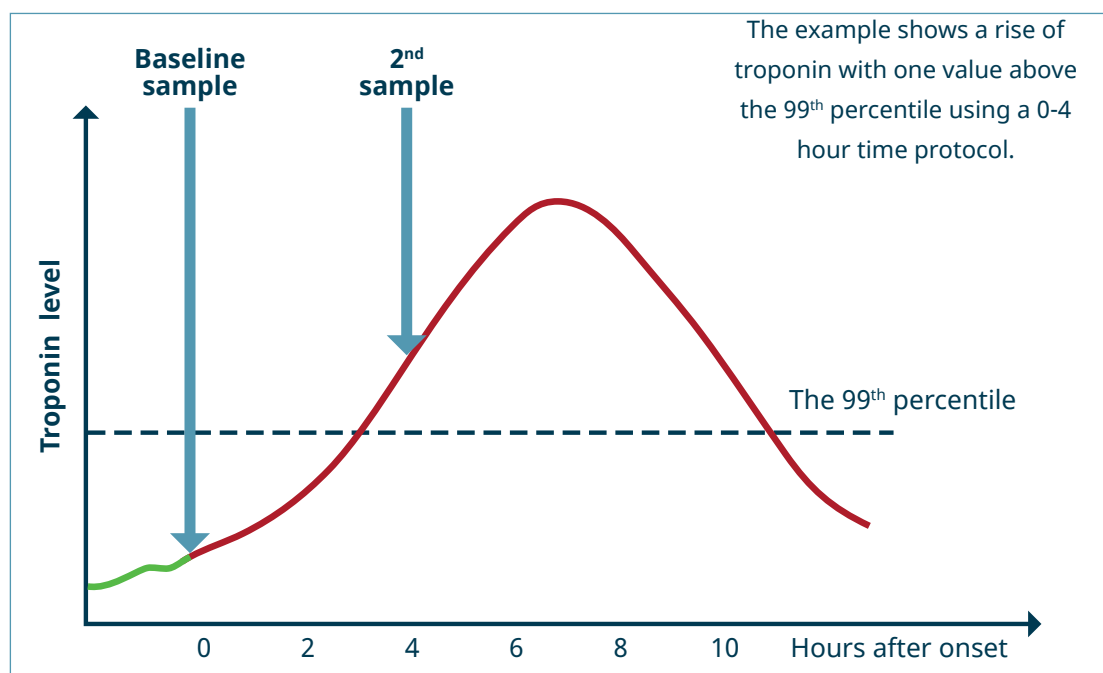


Figure 1. An increase by 50% in cut-off could mean that the patient would not be positive after 4 hours. This illustrates how important the level of the 99th percentile is.

3 The 99th percentile upper reference limit (URL)

Reference population

As illustrated in Figure 1, the level of the 99th percentile is one of the elements that determines how fast a patient can be diagnosed (i.e., the time protocol).

The level of the 99th percentile is determined by conducting a study on an apparently healthy population. The level of the 99th percentile has shown to be dependent on [5]:

- Prescreening of subjects for comorbidities or chronic conditions affecting the heart (e.g., diabetes, renal dysfunction, or heart failure)
- Exclusion of subjects that are prescribed with any cardiovascular medication
- Evaluation of the health history for every subject
- Differences between geographical/ethnic groups

Failing to have met the above criteria, especially if subjects with unknown comorbidities were not excluded, may have led to differences in 99th percentile URL over the years [5, 6].

99th percentile values of the AQT90 FLEX troponin assay

Source	n	99 th percentile URL (µg/L)	CV% at 99 th percentile
Greiser 2017 [7]	2247	0.019	12%
Ivandic 2014 [8]	119	0.020	6.8%
Unpublished internal data 2017 [9]	461	0.014	ND
Unpublished internal data 2008 [10]	231	0.023	12%

ND = not determined

Table 1. The table illustrates differences in 99th percentiles determined on the AQT90 FLEX over the years conducted on different populations and with different exclusion criteria.

4 Performance of troponin

Clinical performance

The clinical performance of a troponin assay is evaluated by a diagnostic accuracy study, providing results for the sensitivity/specificity, negative predictive value (NPV) / positive predictive value (PPV). The diagnostic accuracy at each sample time will define the sample timing protocol and is influenced by:

- The analytical performance of the assay – the analytical sensitivity and the precision profile (CV%)
- The population selected for the clinical study
- The comparator method used to make the original diagnosis
- The 99th percentile

All these elements can vary between studies, and will result in different outcomes (see table 3 on next page).

Analytical performance

The analytical performance of an assay is often described with the assay limit of detection (LOD), the lowest concentration that can be distinguished from a sample containing no troponin and the assay precision at the 99th percentile.

Requirements to CV% [11]

Troponin values at the 99th percentile:

- $CV \leq 10\%$ = “guideline acceptable”
- $CV 10\%-20\%$ = is “clinically usable”
- $CV > 20\%$ = not acceptable

Per definition high-sensitivity troponin assays have CVs < 10% at the 99th percentile.

AQT90 FLEX Precision (CV%) in the literature [8]

Test level (ng/L)	21	20	19	16	11
Total CV (%)	8.5	6.7	8.4	10.5	13.8

Table 2.

↑
99th percentile 2014 [8]
serum pools

Troponin assay precision is inherently dependent on the concentration level.

- A study [8] using 5 different levels below the 99th percentile found CV% well below 10% at the 99th percentile determined in a non-parametric study analyzing 119 healthy subjects in serum pools.
- The 99th percentile was determined to be 20 ng/L and a corresponding CV was found to be 6.7%.
- The Radiometer determined 99th percentile (2008) [10] was found to be 23 ng/L with a CV% in a point-of-care (POC) whole blood study at 10.7%.

Diagnostic accuracy of AQT90 FLEX in the literature

	99 th percentile URL or cut-off used						Diagnostic accuracy						
Source	99 th percentile URL (µg/L)	Method for 99 th percentile URL	CV% at 99 th percentile	n	Population		Population and diagnostic method	n	Sample collection timing	Clinical characteristics			
										Sensitivity	Specificity	NPV	AUC
Ivandic, 2014 [8]	0.020	Non-parametric study	6.7%	119	Healthy individuals		ED patients (cTnT from Roche Cobas e411)	187	0 hours	76%	95%	92%	0.87
									3 hours	88%	94%	96%	0.94
									6 hours	92%	91%	97%	0.95
Suh, 2018 [12]	0.023	Determined by Radiometer	7.4%	-	Healthy individuals		ED patients (hsTnI on Architect i2000SR and hsTnT on Roche Cobas 8000)	76	0 hours	46%	94%	89%	0.84
									3 hours	69%	92%	94%	0.86
Suzuki, 2018 [13]	0.023	Determined by Radiometer	ND	-	Healthy individuals		ED patients	120	< 3 hours	59%	90%	96%	0.80
	0.017 0.063	ROC curve analysis	ND ND	1449	ED patients				> 3 hours	83%	85%	98%	0.89
									< 3 hours	66%	87%	97%	ND
									> 3 hours	79%	92%	98%	ND
Wilke, 2017 [14]	0.023	Determined by Radiometer	12.3%	-	Healthy individuals		ED patients stratified according to renal function (hsTnT and creatinine determined on Roche Cobas 6000)	125	0 hours	79%	95%	ND	0.89
	0.014	ROC curve analysis	ND	573	ED patients without renal and cardiac impairment					83%	90%	ND	
Palamalai, 2013 [15]	0.023	Determined by Radiometer	ND	-	Healthy individuals		ED patients	169	0 hours	26%	93%	91%	0.68
									3 Hours	63%	91%	95%	0.87
									6 Hours	63%	91%	95%	0.92
Schneider, 2013 [16]	0.023	Determined by Radiometer	ND	-	Healthy individuals		ED patients (cTnI on Architect STAT troponin-I assay)	195	0 hours	51%	99%	85%	ND
	0.012	½ x 99 th percentile	ND	-					0 hours	77%	98%	92%	ND
Mion 2016 [17]	0.023	Determined by Radiometer	ND	-	Healthy individuals		ED patients (Diagnosed using Siemens Dimension Vista)	786	0 hours	69%	96%	ND	0.87
	0.020	From Ivandic <i>et al</i> [8]	ND	-	-				0 hours	72%	95%	ND	ND
	0.014	ROC curve analysis	15.3%	786	ED patients				0 hours	79%	93%	ND	ND
Chenevier-Gobeaux, 2018 [18]	0.023	Determined by Radiometer	ND	-	Healthy individuals		ED patients (Multicenter study and multiple methods used for diagnoses)	145	0 hours	33%	95%	91%	
	Delta 40% (H3-H0)	ROC curve analysis	ND	145	ED patients	143		3 hours	67%	92%	95%	0.83	

Table 3. URL = upper reference limit; NPV = negative predictive value; AUC = area under ROC curve; ND = not determined.

Table 3 shows:

99th percentile or cut-off used:

- 99th percentile or cut-off used in clinical study
- If determined, the CV% at the level of the 99th percentile or cut-off used

Diagnostic accuracy:

- **n** shows the number of subjects enrolled for the diagnostic performance study
- The patient population
- The different time protocols used
- The sensitivity/specificity/NPV/AUC of the diagnostic performance studies

Key Takeaways

- The levels of the 99th percentiles show great variance with a tendency to become lower over the years, possibly due to an acknowledgement of the need to exclude certain populations from the reference interval studies. The effect on the clinical characteristics is clear when looking at e.g., the studies by Ivandic [8], Suzuki [13], Wilke [14], Schneider [16], and Mion [17], where the clinical characteristics seem to improve with a lower cut-off.
- The diagnostic accuracy varies with the level of cut-off and sample collection timing. There seems to be a tendency that the clinical characteristics for ruling-in MI patients are not significantly different between 3- and 6-hours sample collection times [8, 15].

5 Turnaround time

Time to result

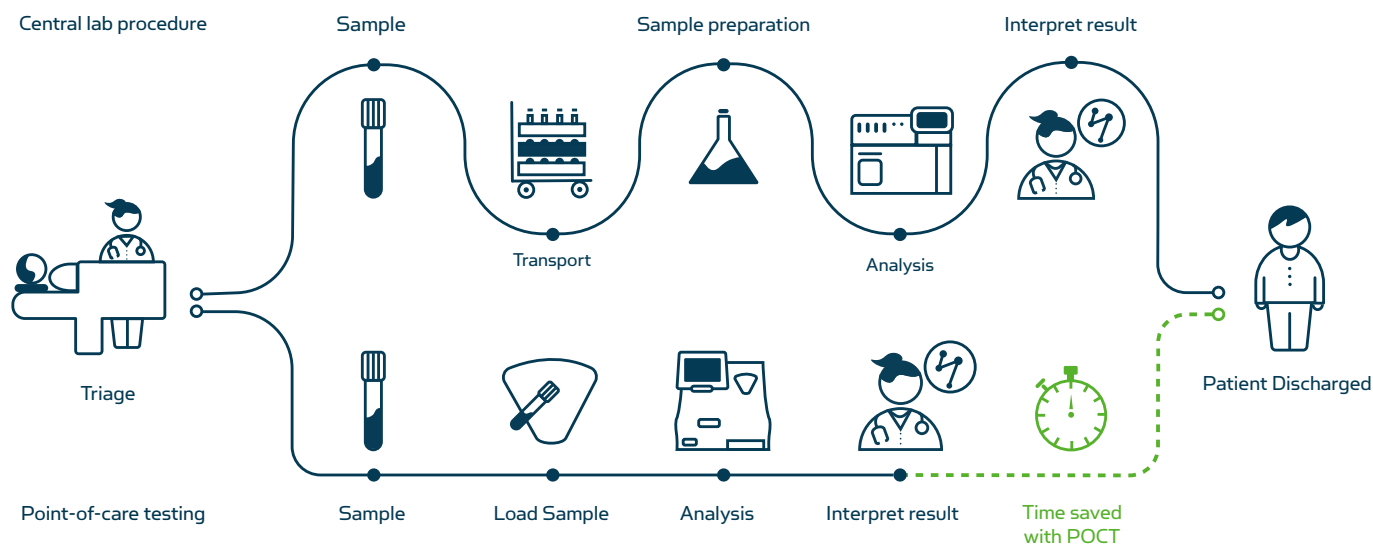
Physicians rely on accurate and timely information to make clinically relevant decisions [19]. Reducing the turnaround time—the time it takes to gather diagnostic information and supply it to the clinician—is paramount in the ED, where a decision-making delay could make a dramatic difference for patients [19, 20, 21].

A recent study [22] found that point-of-care testing (POCT) in an ED had a significant impact on time to result with a reduction of 51 minutes compared to central laboratory testing (CLT).

Length of stay

An earlier randomized controlled trial comparing POC troponin T with CLT using identical decision limits found a significant reduction in length of stay i.e., time from critical care unit admission to hospital discharge or death between the two arms, which were ascribed to the utilization of a structured decision-making protocol [23].

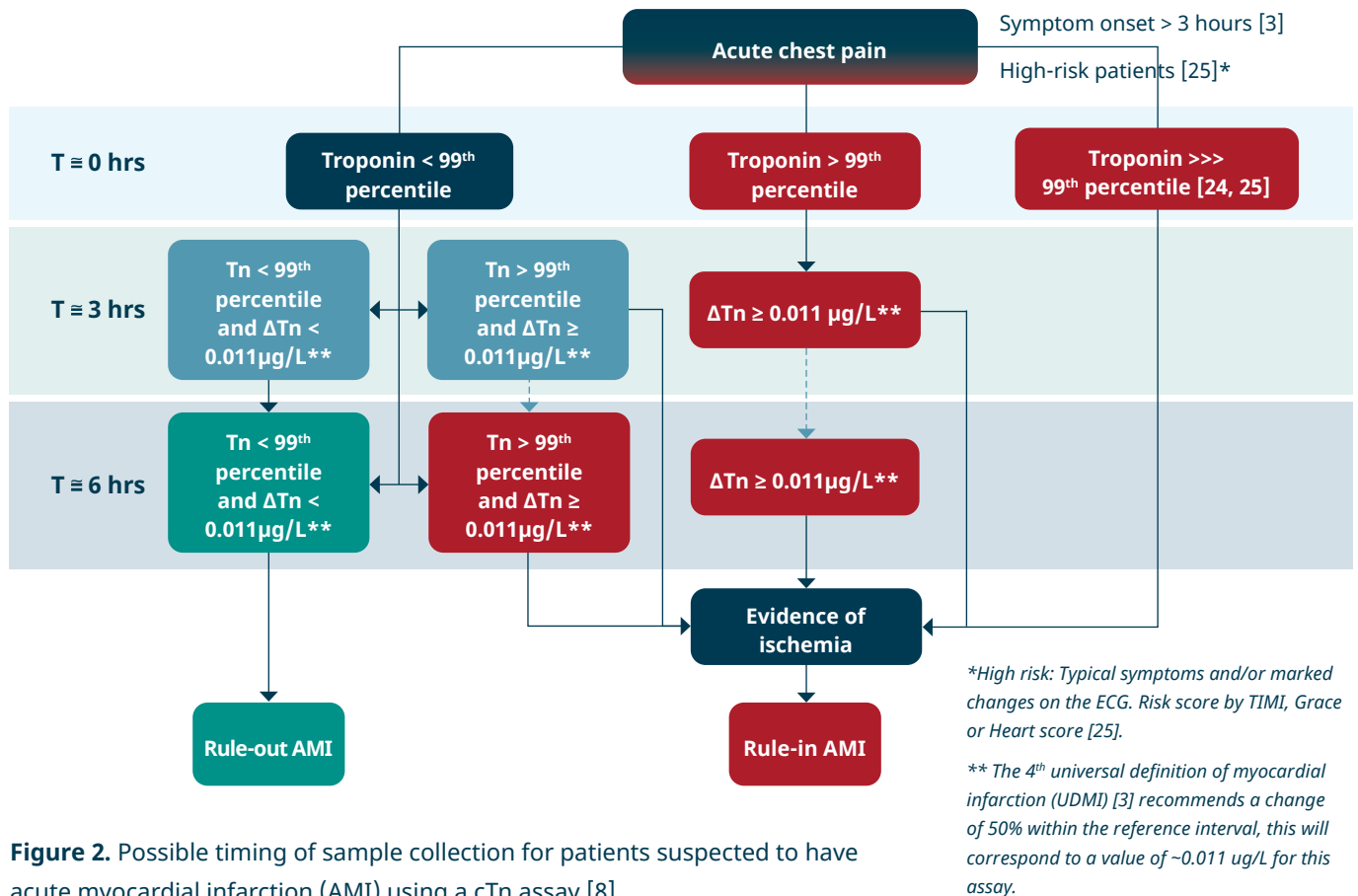
Time saved with point-of-care testing



🔑 POCT has the potential to deliver significant impact on length of stay and ED throughput if it is integrated within the clinical decision-making process [24].

Possible timing of sample collection for patients suspected to have acute myocardial infarction (AMI) using a cTn assay [8]

Results must always be interpreted in the clinical context of the patient.



Timing between sample collection for chest pain patients suspected to have an AMI needs to be considered from patient to patient and depends on both the assay cut-off used and the clinical presentation of the patient. Results must always be interpreted in the clinical context of the patient:

- Early presenters may need a longer time interval between first and second sample to ensure the dynamic release of troponin has brought the concentration above the 99th percentile.
- Late presenters or patients having an extremely high initial troponin measurement may be ruled-in directly on the first sample.
- If there is a dynamic change between two troponin measurements already after 3 hours and sign of ischemia, the patient may already be ruled-in after 3 hours.
- If the change between two serial measurements is not > 0.011 μg/L, then it is suggested to wait another 3 hours for a third sample.



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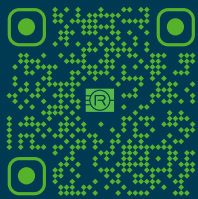
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